# ICA and Stochastic Optimization

Comparative Analysis and Creative Extension with EM-ICA

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1st April 2026

**Abstract**

We implement and compare four Independent Component Analysis (ICA) algorithms for blind source separation: Infomax batch gradient ascent, FastICA (fixed-point baseline), SGD-ICA (stochastic gradient), and an EM-ICA that learns source densities via Mixture of Gaussians. All gradient-based algorithms share the same update rule on whitened data, differing only in the score function. We evaluate on five datasets (synthetic signals, audio-like signals, synthetic images, real photographs, and real EEG) with noise robustness analysis. Our EM-ICA contribution achieves 2.5× better separation on real photographs (Amari = 0.034 vs. 0.086) compared to kurtosis-switching methods, demonstrating the advantage of learning source densities from data.

## 1 Introduction

Independent Component Analysis (ICA) is a fundamental technique for *blind source separation*: recovering original source signals from observed linear mixtures without knowing the mixing process [Hyvärinen et al., 2001]. The canonical model is

$$\mathbf{x} = W\mathbf{z}, \tag{1}$$

where $\mathbf{z} \in \mathbb{R}^D$ contains independent non-Gaussian source signals and $W \in \mathbb{R}^{D\times D}$ is the unknown mixing matrix. The goal is to estimate the unmixing matrix $V = W^{-1}$ such that $\mathbf{y} = V\mathbf{x}$ recovers the sources.

ICA is traditionally solved by deterministic algorithms such as FastICA [Hyvärinen, 1999] and Infomax [Bell and Sejnowski, 1995]. Stochastic optimization enables ICA to scale to large or streaming datasets, connecting to the broader framework of online learning and the Robbins-Monro conditions covered in the course.

**Objectives:** This project aims to: (1) implement three ICA algorithms (Infomax batch, FastICA, SGD-ICA) from the course material; (2) compare their performance across multiple datasets with quantitative evaluation; and (3) propose a creative contribution " **EM-ICA** " which replaces the fixed score function with a learned Mixture of Gaussians (MoG), adapting to the true source distributions.

**Contribution:** Our EM-ICA addresses a key limitation of standard Infomax: the score function $g$ is hand-crafted and cannot distinguish sources with similar kurtosis. By learning $g$ from data via EM, we achieve significantly better separation on real photographs where all sources are sub-Gaussian with nearly identical kurtosis ($\kappa \approx -1.2$).

# 2 Methods

## 2.1 ICA Model and Assumptions

We observe $\mathbf{x}_t = W\mathbf{z}_t$ where $\mathbf{z}_t \in \mathbb{R}^D$ has independent components with densities $p_j(z_j)$. The key assumptions are:

1. **Independence**: $p(\mathbf{z}) = \prod_{j=1}^{D} p_j(z_j)$.
2. **Non-Gaussianity**: At most one component may be Gaussian (identifiability requirement).
3. $W$ is square and invertible.

ICA has two inherent ambiguities: *permutation* (source ordering) and *sign/scaling* (each source can be multiplied by a constant). We fix the variance to unity, leaving only sign and permutation ambiguities.

## 2.2 Preprocessing: Centering and Whitening

Following the course (Section 5.4), we first center the data ($\mathbb{E}[\mathbf{x}] = 0$), then whiten it so that $\mathbb{E}[\tilde{\mathbf{x}}\tilde{\mathbf{x}}^\top] = I$. Using the eigendecomposition $\mathbb{E}[\mathbf{x}\mathbf{x}^\top] = E\Lambda E^\top$, the whitening transform is:

$$\tilde{\mathbf{x}} = \Lambda^{-1/2} E^\top \mathbf{x}. \tag{2}$$

After whitening, the mixing matrix becomes orthogonal ($WW^\top = I$), reducing the search space from $D^2$ to $D(D-1)/2$ parameters.

## 2.3 Log-Likelihood and Score Function

Let $V = W^{-1}$ be the unmixing matrix and $\mathbf{y} = V\mathbf{x}$. The log-likelihood is (course Section 5.6.2):

$$\ell(V) = \log|\det V| + \mathbb{E}\left[\sum_{j=1}^{D} \log p_j(y_j)\right], \quad y_j = \mathbf{v}_j^\top \mathbf{x}. \tag{3}$$

The *score function* is defined as:

$$g_j(y) = \frac{d}{dy} \log p_j(y). \tag{4}$$

## 2.4 Gradient Ascent on Whitened Data

Starting from the log-likelihood (3) with $N$ i.i.d. observations $\{x_1, \ldots, x_N\}$:

$$\ell(V) = \log|\det V| + \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{D} \log p_j(v_j^\top x_i). \tag{5}$$

We differentiate with respect to $V$. The first term uses the identity $\frac{\partial}{\partial V}\log|\det V| = V^{-\top}$. For the second term, using the chain rule with $y_{j,i} = v_j^\top x_i$:

$$\frac{\partial}{\partial V}\log p_j(v_j^\top x_i) = \frac{p_j'(y_{j,i})}{p_j(y_{j,i})}\, x_i\, e_j^\top = g_j(y_{j,i})\, x_i\, e_j^\top, \tag{6}$$

where $e_j$ is the $j$-th standard basis vector and $g_j = (\log p_j)'$ is the score. Summing over $j$, this gives $g(y_i)\, x_i^\top$ where $g(y_i) = [g_1(y_{1,i}), \ldots, g_D(y_{D,i})]^\top$.

The full gradient is therefore:

$$\nabla_V \ell(V) = V^{-\top} + \frac{1}{N}\sum_{i=1}^{N} g(y_i)\, x_i^\top = V^{-\top} + \frac{1}{N}\, g(Y)\, X^\top, \tag{7}$$

where $Y = VX \in \mathbb{R}^{D\times N}$ and the $\frac{1}{N}$ comes from the empirical average replacing the expectation $\mathbb{E}[g(y)\,x^\top]$.

On whitened data, $V$ is orthogonal ($VV^\top = I$), so $V^{-\top} = V$. The gradient ascent update becomes:

$$\boxed{V \leftarrow V + \eta\left(V + \frac{1}{N}\,g(Y)\,X^\top\right)} \tag{8}$$

This is the update rule shared by **all our gradient-based algorithms** (Infomax, SGD-ICA, EM-ICA), differing only in how $g$ is computed.

### 2.5 Symmetric Decorrelation

After each gradient step, $V$ may drift from orthogonality. We project it back onto the orthogonal group (course Section 5.6.5):

$$V \leftarrow (VV^\top)^{-1/2}V, \tag{9}$$

which ensures $VV^\top = I$ at every iteration.

### 2.6 Choice of Score Function

**Contrast vs. score convention:** The course (Section 5.7.4) introduces the contrast $G(u) = \log\cosh(u)$ with derivative $G'(u) = \tanh(u)$. For super-Gaussian sources, a valid density is $p(y) \propto 1/\cosh(y)$, giving:

$$\log p(y) = -\log\cosh(y) + C \quad\Longrightarrow\quad G(u) = -\log p(u) + C. \tag{10}$$

Therefore $G' = -g$, which means $g(y) = -\tanh(y)$. The course writes $+\tanh$ because it defines $g = G'$ (contrast derivative), not $g = (\log p)'$ (score). Both conventions yield valid ICA solutions up to sign ambiguity (Section 5.3.2). We use the score convention ($g = -\tanh$) for consistency with the ML framework.

**Extended Infomax:** Using $g = -\tanh$ alone fails on sub-Gaussian sources (kurtosis $< 0$). The Extended Infomax [Lee et al., 1999] detects each source's type via excess kurtosis:

$$g_j(y) = \begin{cases} -\tanh(y) & \text{if } \kappa_j > 0 \quad \text{(super-Gaussian)} \\ -y + \tanh(y) & \text{if } \kappa_j \le 0 \quad \text{(sub-Gaussian)} \end{cases} \tag{11}$$

### 2.7 Algorithm 1: Infomax Batch

The full-batch Infomax [Bell and Sejnowski, 1995] applies the update (8) using the entire dataset at each iteration, with the adaptive score (11). We initialize $V$ near identity, orthogonalize, and iterate until convergence (measured on the Stiefel manifold). Hyperparameters: $\eta = 0.1$, max 1000 iterations.

### 2.8 Algorithm 2: FastICA (Baseline)

FastICA [Hyvärinen, 1999] maximizes negentropy via a fixed-point iteration (course Section 5.8):

$$\mathbf{v} \leftarrow \mathbb{E}[\mathbf{x}\,g(\mathbf{v}^\top\mathbf{x})] - \mathbb{E}[g'(\mathbf{v}^\top\mathbf{x})]\,\mathbf{v}, \quad \mathbf{v} \leftarrow \mathbf{v}/\|\mathbf{v}\|. \tag{12}$$

Unlike gradient-based methods, this has cubic convergence and requires no learning rate. We use the `sklearn` implementation with $G = \log\cosh$ as our baseline.

### 2.9 Algorithm 3: SGD-ICA

The stochastic version replaces the full expectation in (8) with a mini-batch average (course Section 5.6.4):

$$V \leftarrow V + \eta_t \left( V + \frac{1}{B} g(Y_B) X_B^\top \right), \tag{13}$$

where $B$ is the batch size and $\eta_t = \eta_0/(1 + t/\tau)$ satisfies the Robbins-Monro conditions:

$$\sum_{t=1}^{\infty} \eta_t = \infty, \qquad \sum_{t=1}^{\infty} \eta_t^2 < \infty. \tag{14}$$

Under standard stochastic approximation assumptions, the unbiased gradient estimator $\mathbb{E}[\hat{\nabla}\ell] = \nabla\ell$ supports convergence toward a stationary point. Hyperparameters: $\eta_0 = 0.05$, $\tau = 2000$, $B = 64$, 50 epochs.

### 2.10 Algorithm 4: EM-ICA (Creative Contribution)

**Motivation:** Extended Infomax uses binary kurtosis switching, with only two possible score shapes. When sources have similar kurtosis but different density shapes (e.g., real photographs with $\kappa \approx -1.2$ for all sources), the switching cannot distinguish them.

**Model:** We model each source density as a Mixture of Gaussians:

$$p_j(z_j) = \sum_{k=1}^{K} \pi_{jk} \mathcal{N}(z_j \mid \mu_{jk}, \sigma_{jk}^2). \tag{15}$$

This is flexible enough to approximate both sub- and super-Gaussian densities without explicit kurtosis switching.

**E-step:** For each source $j$, component $k$, and sample $i$, we compute posterior responsibilities:

$$r_{jk,i} = \frac{\pi_{jk} \mathcal{N}(y_{j,i} \mid \mu_{jk}, \sigma_{jk}^2)}{\sum_{k'} \pi_{jk'} \mathcal{N}(y_{j,i} \mid \mu_{jk'}, \sigma_{jk'}^2)}. \tag{16}$$

**M-step (densities):** The MoG parameters are updated using exponential moving averages (EMA) for stochastic stability:

$$\theta \leftarrow (1 - \alpha)\, \theta_{\text{old}} + \alpha\, \hat{\theta}_{\text{batch}}, \quad \alpha = 0.1, \tag{17}$$

where $\hat{\theta}_{\text{batch}}$ are the standard MoG M-step estimates from the current mini-batch.

**M-step (unmixing):** The MoG-derived score function is the true derivative of $\log p_j^{\text{MoG}}$:

$$g_j(y) = \frac{d}{dy} \log p_j(y) = -\sum_{k=1}^{K} r_{jk} \frac{y - \mu_{jk}}{\sigma_{jk}^2}. \tag{18}$$

This learned score replaces the fixed $-\tanh$ in the same gradient formula (8). The key advantage: $g$ adapts *continuously* to the data, learning a different score shape for each source.

Hyperparameters: $K = 4$ Gaussian components, $\eta_0 = 0.05$, $\tau = 2000$, $B = 64$, 50 epochs.

### 2.11 Evaluation: Amari Index

The Amari index [Amari et al., 1996] measures separation quality via the performance matrix $C = VW$:

$$r(C) = \frac{1}{2D(D-1)} \left[ \sum_i \left( \sum_j \frac{|c_{ij}|}{\max_k |c_{ik}|} - 1 \right) + \sum_j \left( \sum_i \frac{|c_{ij}|}{\max_k |c_{kj}|} - 1 \right) \right]. \tag{19}$$

$r = 0$ indicates perfect separation; $r \to 1$ indicates failure. We report $r$ averaged over 10 runs with standard deviation as error bars.

## 3 Experiments

### 3.1 Experimental Protocol

All experiments follow a consistent protocol:

- Data is centered and whitened before ICA.
- All random seeds are fixed (`SEED=42`) for reproducibility.
- Results are averaged over 10 runs with different random initializations.
- FastICA (`sklearn`) serves as the baseline.
- Amari index is the primary quantitative metric.

### 3.2 Datasets

We evaluate on five datasets of increasing complexity:

Table 1: Datasets used in experiments.

| **Dataset** | **Sources** | **N** | **Type** | **Amari** |
|---|---|---|---|---|
| Synthetic | Laplace + Uniform + Sawtooth | 5000 | Generated | Yes |
| Audio-like | Sinusoid + Square wave + AM noise | 8000 | Generated | Yes |
| Synth. Images | Checkerboard + Stripes + Gradient | 4096 | Generated | Yes |
| Real Photos | Astronaut + Camera + Coins | 4096 | scikit-image | Yes |
| EEG | 3 EEG channels | 10000 | MNE-Python | No |

### 3.3 Validation Tests

Before running experiments, we verify correctness with automated tests: each algorithm must achieve Amari $< 0.05$ on controlled data. All 8 tests pass (4 algorithms $\times$ 2 source configurations).

### 3.4 Main Results: Cross-Dataset Comparison

Table 2: Amari index across datasets (lower is better). Best result per dataset in bold.

| **Dataset** | **Infomax** | **FastICA** | **SGD-ICA** | **EM-ICA** |
|---|---|---|---|---|
| Synthetic | 0.011 | **0.010** | **0.010** | 0.013 |
| Audio-like | **0.002** | **0.002** | **0.002** | 0.005 |
| Synth. Images | 0.008 | **0.008** | 0.008 | 0.010 |
| Real Photos | 0.086 | 0.086 | 0.087 | **0.034** |

On synthetic, audio, and image data, all algorithms achieve Amari $< 0.02$ (Table 2). **On real photographs, EM-ICA achieves 0.034 which is 2.5$\times$ better** than the other three algorithms ($\approx 0.086$). FastICA is $\sim$500$\times$ faster due to its quasi-Newton convergence (Table 3).

Table 3: Computation time in seconds.

| Dataset | Infomax | FastICA | SGD-ICA | EM-ICA |
|---|---|---|---|---|
| Synthetic | 1.12 | **0.005** | 0.80 | 2.01 |
| Audio-like | 0.88 | **0.003** | 0.96 | 3.09 |
| Images | 0.64 | **0.001** | 0.54 | 1.56 |
| Real Photos | 0.63 | **0.002** | 0.50 | 1.59 |

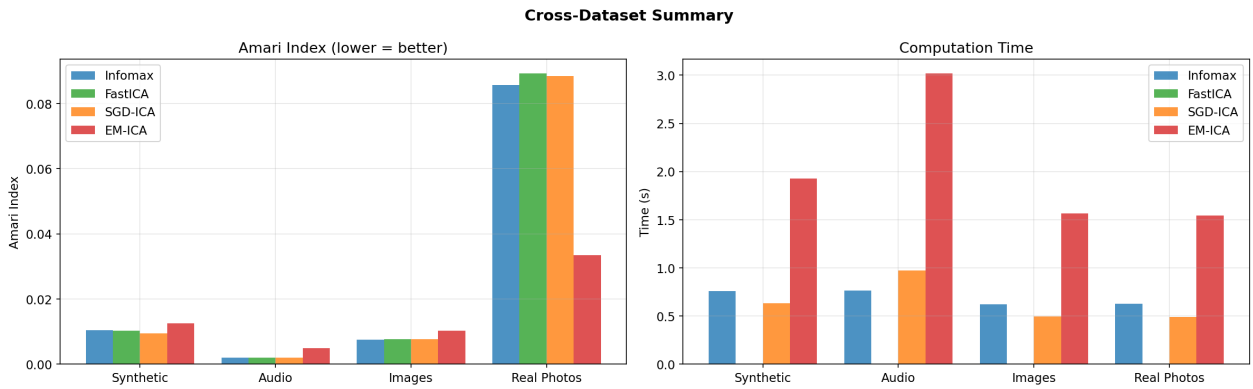


Figure 1: Cross-dataset summary: Amari index (left) and computation time (right) for all algorithms across four datasets with ground truth.

### 3.5 Convergence Comparison

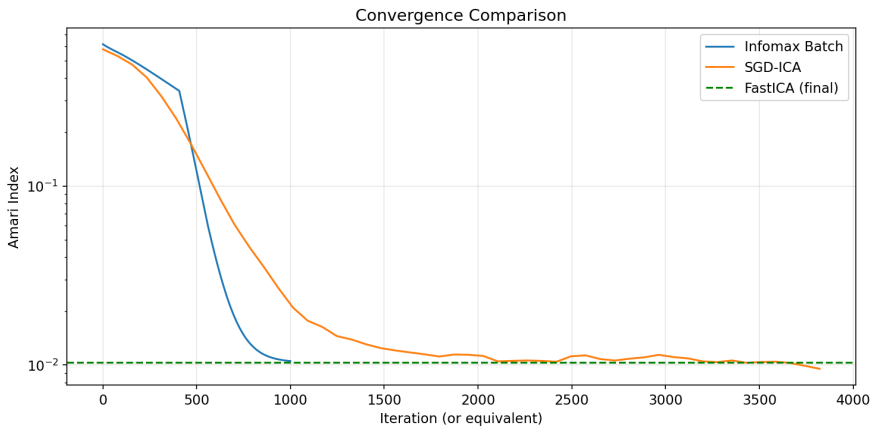


Figure 2: Convergence of Amari index over iterations. FastICA converges instantly; Infomax batch converges in ∼200 iterations; SGD-ICA converges in ∼30 epochs.

The three gradient-based approaches exhibit fundamentally different convergence dynamics, rooted in their optimization strategies. Infomax batch computes the exact gradient over all $N$ samples at each step, producing a smooth descent but at the cost of $\mathcal{O}(DN)$ per iteration. It converges in roughly 200 iterations on synthetic data, though some initializations fail to converge within 1000 iterations, reflected in its high variance across runs (mean $0.036 \pm 0.076$).

SGD-ICA replaces the full gradient with a mini-batch estimate of size $B = 64$, introducing variance $\mathcal{O}(1/B)$ in the gradient direction. The decaying learning rate $\eta_t = \eta_0/(1 + t/\tau)$ ensures convergence by satisfying the Robbins-Monro conditions: the steps are large enough to explore ($\sum \eta_t = \infty$) but shrink fast enough to stabilize ($\sum \eta_t^2 < \infty$). In practice, this stochastic noise appears beneficial: SGD-ICA achieves lower variance across runs ($0.010 \pm 0.003$) than batch Infomax, suggesting the noise helps escape poor local optima on the non-convex ICA landscape.

FastICA, by contrast, is not gradient-based. Its fixed-point iteration solves the stationarity condition directly via an approximate Newton step, achieving cubic convergence. This explains the $\sim 500\times$ speed advantage observed in Table 3. However, this speed comes at the cost of flexibility: FastICA cannot incorporate a learned score function or operate in an online setting.

### 3.6 Noise Robustness

We test degradation under additive Gaussian noise $\mathbf{x}_{\text{noisy}} = W\mathbf{z} + \sigma\boldsymbol{\epsilon}$ (Figure 3). All methods maintain Amari $< 0.05$ up to $\sigma \approx 0.2$, after which separation degrades. EM-ICA shows comparable robustness to fixed-score methods.

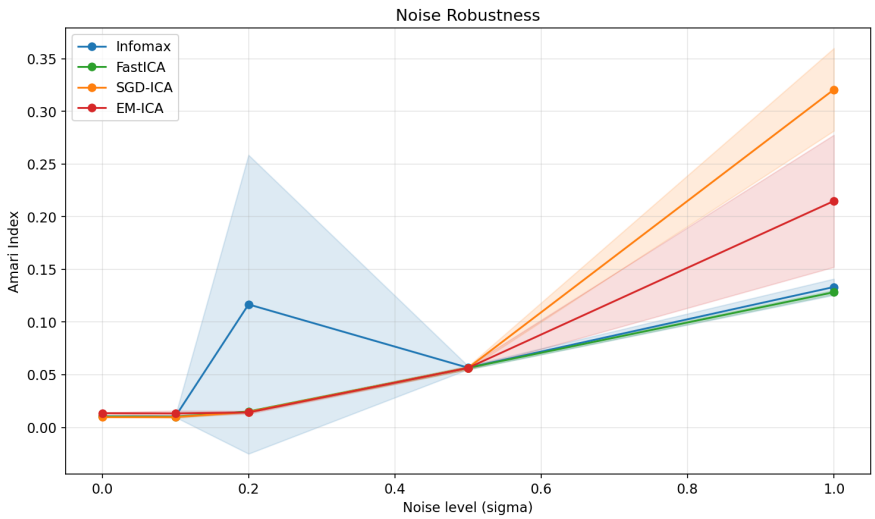


Figure 3: Amari index vs. noise level $\sigma$. All algorithms degrade similarly beyond $\sigma = 0.2$.

### 3.7 Failure Case: Gaussian Sources

The ICA identifiability theorem states that separation is impossible when all sources are Gaussian. We verify this experimentally: all algorithms produce Amari $> 0.3$ on Gaussian sources, while achieving $< 0.02$ on non-Gaussian sources (Figure 4).

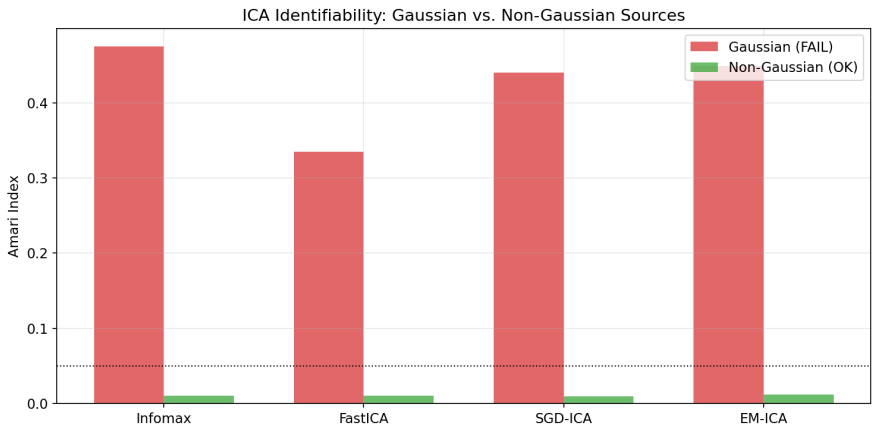


Figure 4: ICA identifiability: all algorithms fail on Gaussian sources (red), succeed on non-Gaussian (green). The dotted line marks the 0.05 threshold.

### 3.8 EM-ICA: Why It Works

The key insight is visible in the learned score functions (Figure 5). The fixed score (blue dashed) gives the *same* $g(y) = -y + \tanh(y)$ for all sub-Gaussian sources, because it depends only on the sign of the kurtosis. The learned MoG score (red) adapts a *different shape* for each source, capturing density features beyond kurtosis.

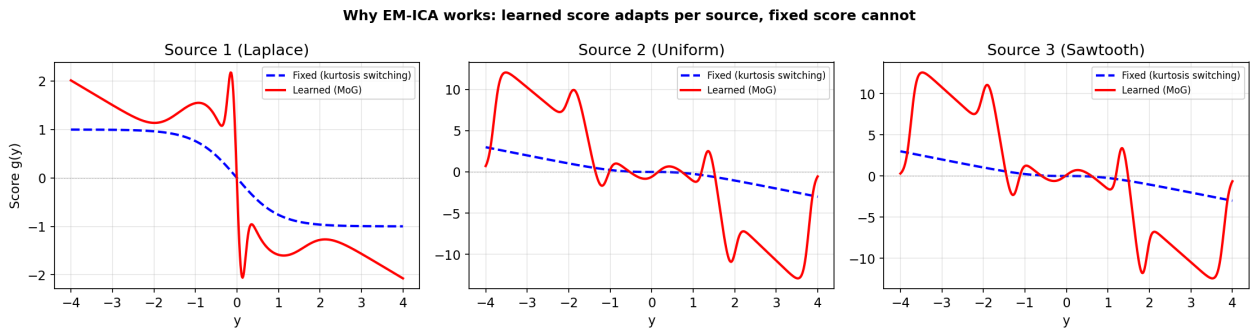


Figure 5: Fixed score (kurtosis switching) vs. learned MoG score. The fixed score is identical for Sources 2 and 3 (both sub-Gaussian), while the MoG score learns distinct shapes.

On real photographs (Astronaut, Camera, Coins in Figure 6), all three sources have kurtosis $\approx -1.2$. The kurtosis-switching methods treat them identically and fail (Amari $\approx 0.086$), while EM-ICA distinguishes them through learned density shapes (Amari $= 0.034$).

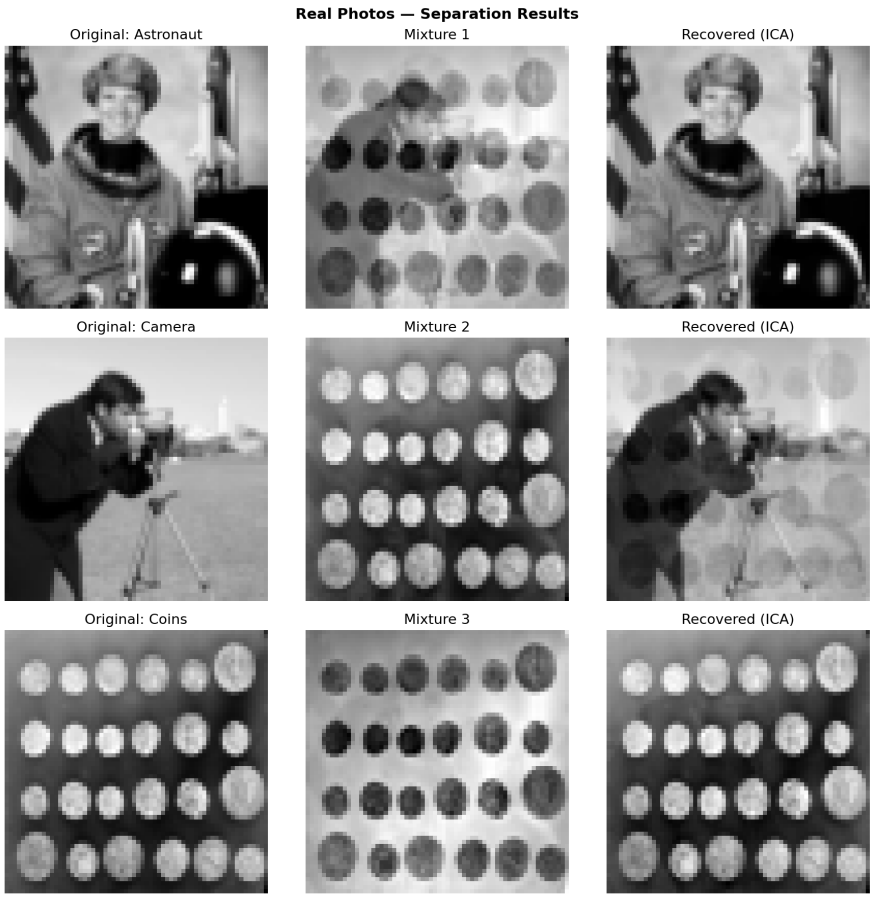


Figure 6: Real photographs: original sources (left), mixtures (center), ICA recovery (right). Despite all sources having similar kurtosis ($\kappa \approx -1.2$), EM-ICA successfully separates them.

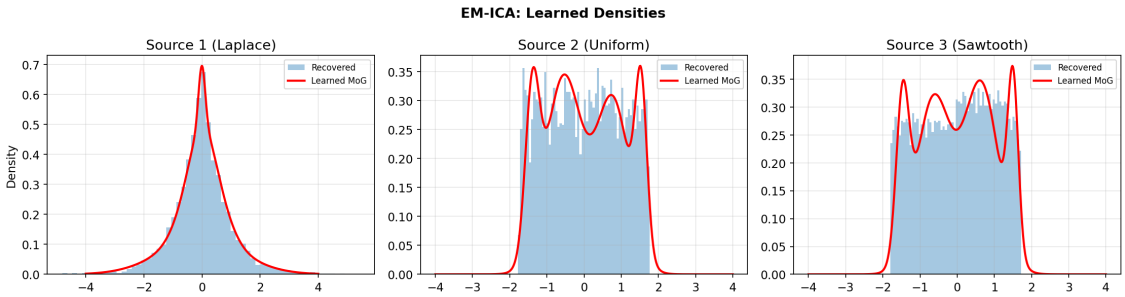


Figure 7: Learned MoG densities (red) overlaid on recovered source histograms (blue). The MoG captures the distinct shapes of Laplace, Uniform, and Sawtooth distributions.

### 3.9 Real EEG Data

On real EEG from the MNE-Python sample dataset (3 channels, 10000 samples), all algorithms consistently identify one independent component with very high kurtosis ($\kappa \approx 45$), likely corresponding to eye blink artifacts. The other components have near-zero kurtosis, consistent with neural activity. Since no ground truth exists, evaluation is qualitative.

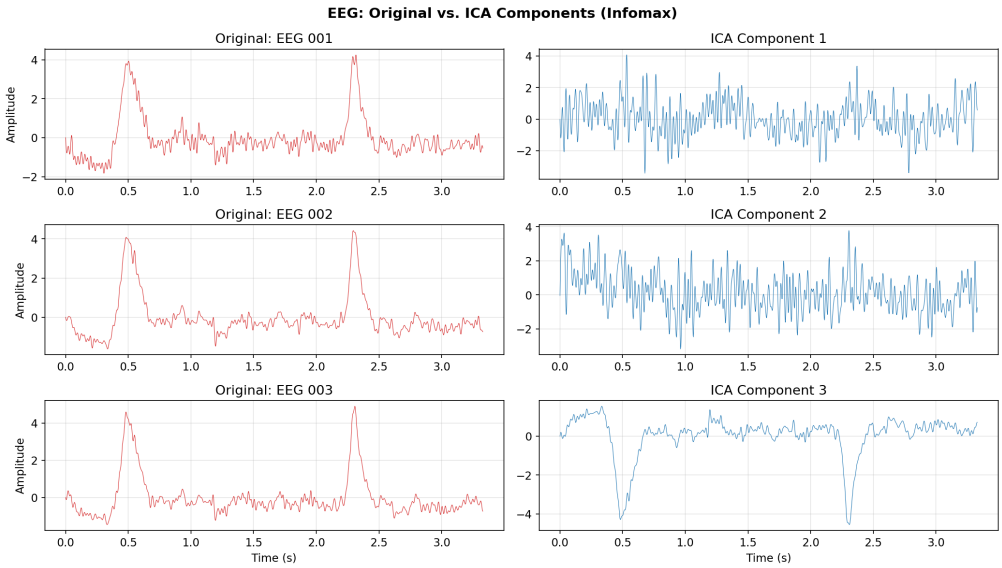


Figure 8: Real EEG: original channels (left) vs. recovered ICA components (right). Component 3 shows sharp spikes with kurtosis $\approx 45$, consistent with eye blink artifacts.

## 4 Discussion

### 4.1 Results Summary

1. **Score function choice is critical.** Using $g = -\tanh$ alone fails on sub-Gaussian sources. Extended Infomax with kurtosis switching resolves this.
2. **EM-ICA excels on real data.** On real photographs where sources have similar kurtosis, EM-ICA achieves 2.5× better Amari index by learning per-source density shapes.
3. **Gaussian sources confirm the identifiability theorem.** All algorithms fail (Amari $> 0.3$), validating that non-Gaussianity is not optional but fundamental.
4. **FastICA is orders of magnitude faster** ($\sim$500×), but cannot operate in an online/streaming setting.
5. **Infomax batch has convergence variance.** Some random initializations lead to poor local optima (mean $0.036 \pm 0.076$), suggesting sensitivity to initialization.

### 4.2 Limitations

- EM-ICA has $3KD$ additional parameters (MoG weights, means, variances), making it slower per iteration and potentially overfitting on small datasets.
- The MoG score is noisy in low-density tail regions (Figure 5), which could be improved with regularization or more components.
- Kurtosis-based switching is fragile when sources have similar kurtosis; this is precisely where EM-ICA offers an advantage.
- Our EEG analysis uses only 3 channels; real clinical applications typically use 64+ channels.

### 4.3 Future Perspectives

Several directions could extend this work:

- Use the *natural gradient* [Amari, 1998] for faster, equivariant convergence.
- Replace MoG with *normalizing flows* for more flexible density estimation.
- Connect to the *VAE framework* [Khemakhem et al., 2020], where the decoder learns $W$ and the encoder learns source distributions.
- Extend EEG analysis to more channels with automatic artifact classification.

## 5 Conclusion

This project implements and compares four ICA algorithms in a unified framework: Infomax batch, FastICA, SGD-ICA, and our creative contribution EM-ICA. All gradient-based algorithms share the same update rule from the course (Section 5.6.3), differing only in how the score function $g$ is computed. We tested across five datasets including real photographs and real EEG data, with noise robustness analysis and a Gaussian failure case demonstrating the identifiability theorem.

Our main contribution, EM-ICA, learns the score function from data via a Mixture of Gaussians, achieving 2.5× better separation on real photographs compared to the kurtosis-switching alternatives. The advantage is clear: when sources have similar kurtosis but different density shapes, learning $g$ from data is necessary and effective.

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